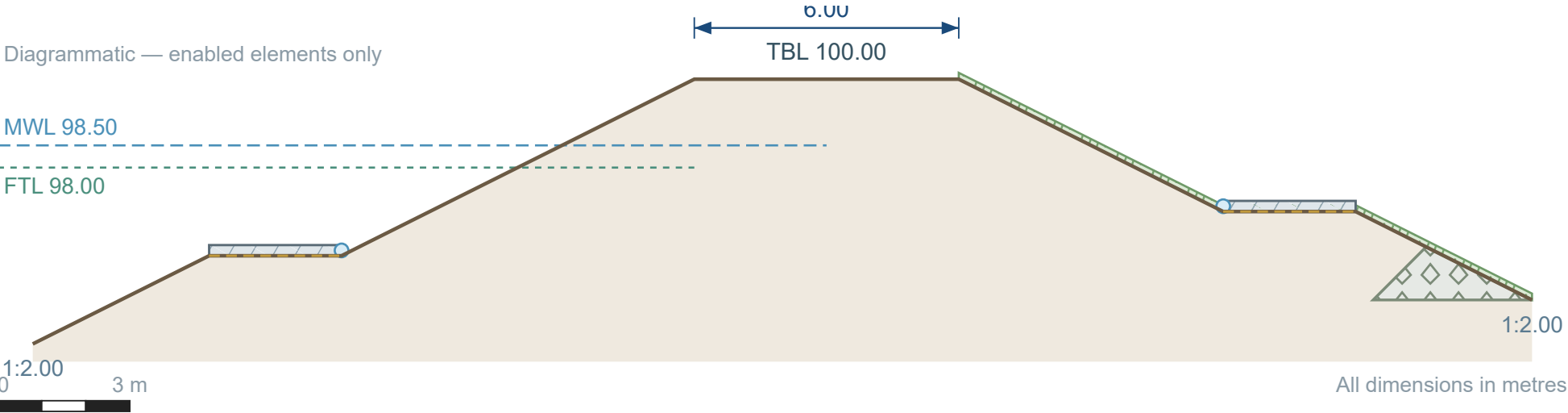


Bund #1 [West]

NEW · HOMOGENEOUS

Test reservoir · 3 surveyed section(s) · Bund length 60 m · Datum 0 m · Crest width 6 m

MWL (m)	Freeboard (m)	TBL (m)	FTL (m)	U/S slope	D/S slope
98.5	1.5	100	98	2:1	2:1



Statement of quantities

Average section value = (A1 + A2) / 2. Interval quantity = average section value × length.

Chainage	Length (m)	Average toe RL (m)	Bund height (m)	Bund width / surveyed ground perimeter (m)	Jungle clearance			Bund foundation excavation — depth 0.3 m			Homogeneous bund filling (excluding berms)		
					Section	Average	Total	Section	Average	Total	Section	Average	Total
					(m)	(m)	(m²)	(m²)	(m²)	(m³)	(m²)	(m²)	(m³)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
0	—	94.5*	5.8	34.003	34	—	—	10.2	—	—	91.29	—	—
30	30	95*	5.3	32.002	32	33	990	9.6	9.9	297	88.728	90.009	2700.27
60	30	95	5.3	32	32	32	960	9.6	9.6	288	77.84	83.284	2498.52
Total	60						1950			585			5198.79

Statement of quantities — continued

Chainage	Length (m)	D/S turfing			Rock toe			Rock-toe foundation excavation			Rock-toe graded filter		
		Section	Average	Total	Section	Average	Total	Section	Average	Total	Section	Average	Total
		(m)	(m)	(m ²)	(m ²)	(m ²)	(m ³)	(m ²)	(m ²)	(m ³)	(m ²)	(m ²)	(m ³)
(1)	(2)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)	(26)
0	—	8.497	—	—	2.16	—	—	0	—	—	4.449	—	—
30	30	8.497	8.497	254.91	2.16	2.16	64.8	0	0	0	4.449	4.449	133.47
60	30	8.497	8.497	254.91	2.16	2.16	64.8	0	0	0	4.449	4.449	133.47
Total	60			509.82			129.6			0			266.94

Statement of quantities — continued

Chainage	Length (m)	Chute-drain excavation			Chute-drain lining			Berm filling (U/S, RL 96)			Berm protection (U/S, RL 96)		
		Section	Average	Total	Section	Average	Total	Section	Average	Total	Section	Average	Total
		(m ²)	(m ²)	(m ³)	(m ³ /m)	(m ³ /m)	(m ³)	(m ²)	(m ²)	(m ³)	(m)	(m)	(m ²)
(1)	(2)	(27)	(28)	(29)	(30)	(31)	(32)	(33)	(34)	(35)	(36)	(37)	(38)
0	—	0.067	—	—	0.045	—	—	5.096	—	—	2.4	—	—
30	30	0.067	0.067	2.012	0.045	0.045	1.342	3.956	4.526	135.78	2.4	2.4	72
60	30	0.067	0.067	2.012	0.045	0.045	1.342	3	3.478	104.34	2.4	2.4	72
Total	60			4.024			2.684			240.12			144

Statement of quantities — continued

Chainage	Length (m)	Berm drain protection (U/S, RL 96)			Berm drain excavation (U/S, RL 96)			Berm filling (D/S, RL 97)			Berm protection (D/S, RL 97)		
		Section	Average	Total	Section	Average	Total	Section	Average	Total	Section	Average	Total
		(m)	(m)	(m ³)	(m ²)	(m ²)	(m ³)	(m ²)	(m ²)	(m ³)	(m)	(m)	(m ²)
(1)	(2)	(39)	(40)	(41)	(42)	(43)	(44)	(45)	(46)	(47)	(48)	(49)	(50)
0	—	0.12	—	—	0.18	—	—	6.957	—	—	2.4	—	—
30	30	0.12	0.12	3.6	0.18	0.18	5.4	6.956	6.957	208.71	2.4	2.4	72
60	30	0.12	0.12	3.6	0.18	0.18	5.4	6	6.478	194.34	2.4	2.4	72
Total	60			7.2			10.8			403.05			144

Statement of quantities — continued

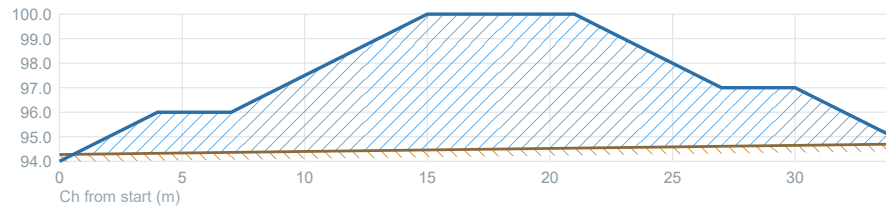
Chainage	Length (m)	Berm drain protection (D/S, RL 97)			Berm drain excavation (D/S, RL 97)		
		Section	Average	Total	Section	Average	Total
		(m)	(m)	(m ³)	(m ²)	(m ²)	(m ³)
(1)	(2)	(51)	(52)	(53)	(54)	(55)	(56)
0	—	0.12	—	—	0.18	—	—
30	30	0.12	0.12	3.6	0.18	0.18	5.4
60	30	0.12	0.12	3.6	0.18	0.18	5.4
Total	60			7.2			10.8

* Detailed existing-ground cross-section; bund width is the surveyed ground perimeter between the designed toes.

Cross-sections

Ch 0 · Average toe RL 94.5 m

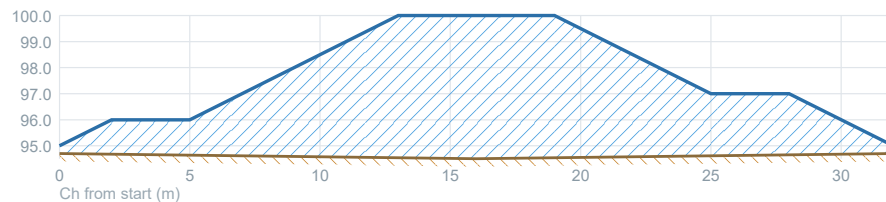
SURVEYED GROUND PERIMETER	FOUNDATION / STRIPPING AREA
34.003 m	10.2 m ²



Ch	EL	RL	Calculation	Quantity
0	94.275	94	Start point	—
0.564	94.282	94.282	$0.564 \times [(-0.275) + (0)] / 2$	-0.078
4	94.325	96	$3.436 \times [(0) + (1.675)] / 2$	2.878
7	94.363	96	$3 \times [(1.675) + (1.637)] / 2$	4.969
7.5	94.369	96.25	$0.5 \times [(1.637) + (1.881)] / 2$	0.88
15	94.463	100	$7.5 \times [(1.881) + (5.537)] / 2$	27.82
18	94.5	100	$3 \times [(5.537) + (5.5)] / 2$	16.556
21	94.538	100	$3 \times [(5.5) + (5.462)] / 2$	16.444
27	94.613	97	$6 \times [(5.462) + (2.388)] / 2$	23.55
27.5	94.619	97	$0.5 \times [(2.388) + (2.381)] / 2$	1.192
30	94.65	97	$2.5 \times [(2.381) + (2.35)] / 2$	5.914
34	94.7	95	$4 \times [(2.35) + (0.3)] / 2$	5.3
Total formation quantity				105.425 m ²

Ch 30 · Average toe RL 95 m

SURVEYED GROUND PERIMETER	FOUNDATION / STRIPPING AREA
32.002 m	9.6 m ²



Ch	EL	RL	Calculation	Quantity
0	94.7	95	Start point	—
2	94.675	96	$2 \times [(0.3) + (1.325)] / 2$	1.625
5	94.638	96	$3 \times [(1.325) + (1.362)] / 2$	4.031
6.5	94.619	96.75	$1.5 \times [(1.362) + (2.131)] / 2$	2.62
13	94.538	100	$6.5 \times [(2.131) + (5.463)] / 2$	24.68
16	94.5	100	$3 \times [(5.463) + (5.5)] / 2$	16.444
19	94.538	100	$3 \times [(5.5) + (5.463)] / 2$	16.444
25	94.613	97	$6 \times [(5.463) + (2.388)] / 2$	23.55
25.5	94.619	97	$0.5 \times [(2.388) + (2.381)] / 2$	1.192
28	94.65	97	$2.5 \times [(2.381) + (2.35)] / 2$	5.914
32	94.7	95	$4 \times [(2.35) + (0.3)] / 2$	5.3
Total formation quantity				101.8 m ²

Earthwork excavation classification

Percentages are the selected dashboard classification applied to each measured source.

Foundation / stripping · 585 m³

Soil / rock	Share (%)	Code	Calculation	Quantity (m ³)
All Soils	100	IRR-CAW-1-1	$585 \times 100 / 100$	585

Chute drain · 4.024 m³

Soil / rock	Share (%)	Code	Calculation	Quantity (m ³)
All Soils	100	IRR-CAW-1-1	$4.024 \times 100 / 100$	4.024

Berm catch-water drains · 21.6 m³

Soil / rock	Share (%)	Code	Calculation	Quantity (m ³)
All Soils	100	IRR-CAW-1-1	$21.6 \times 100 / 100$	21.6

Total quantities of Excavation.

Each excavation DATA code is shown once, followed by the works contributing to its total.

IRR-CAW-1-1.

Foundation stripping	Chute drains	Berm catch-water drains	Total
585	+ 4.024	+ 21.6	= <u>610.624</u> m ³

Total quantities

Each work code is shown once, followed by the components that add to its total. Excavation quantities are in Total quantities of Excavation above and are not repeated here. Distinct operations can share a measured volume; their quantities must not be added to obtain geometric fill.

IRR-PMW-1-2.

Jungle clearance	Total
1950	= <u>1950</u> m ²

IRR-DAW-5-6.

Bund	u/s Berm (RL 96)	d/s Berm (RL 97)	Less: Rock toe	Total
5328.39	+ 240.12	+ 403.05	-129.6	= <u>5841.96</u> m ³

IRR-DAW-6-15.

Bund	Total
509.82	= <u>509.82</u> SQM

IRR-CAW-7-15.

D/S chute drains — channel protection	Total
2.684	= <u>2.684</u> CUM

IRR-CAW-8-15.

u/s Berm (RL 96)	Total
288	= <u>288</u> SQM

IRR-CAW-7-12.

u/s Berm (RL 96)	Total
14.4	= <u>14.4</u> CUM

IRR-CAW-5-6.

Rock toe (d/s)	Total
129.6	= <u>129.6</u> CUM

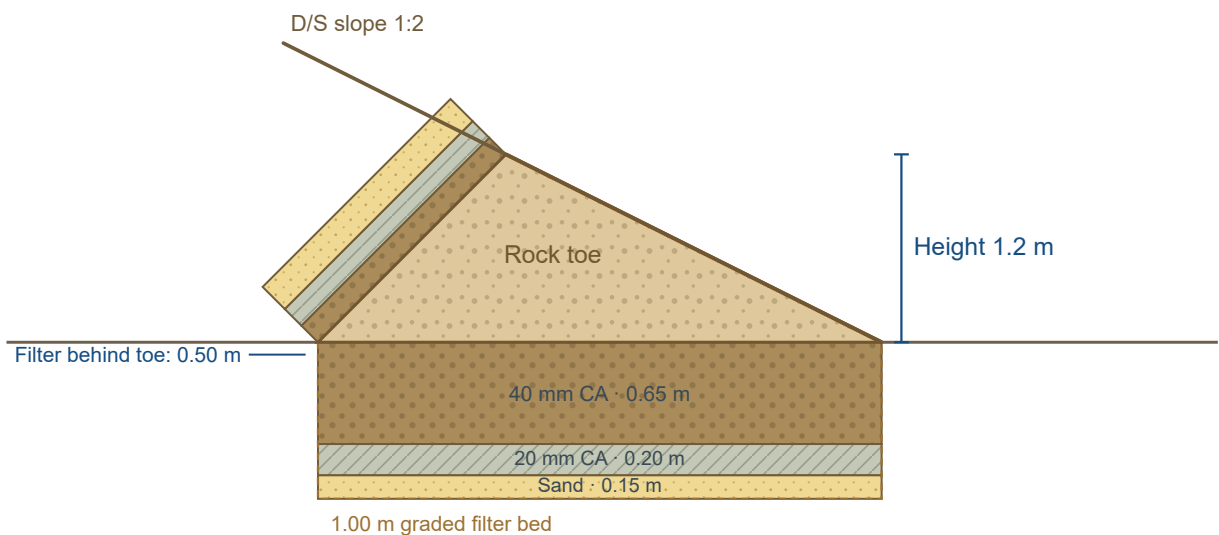
IRR-CAW-5-11.

Graded filter below and behind rock toe (d/s)	Total
266.94	= <u>266.94</u> CUM

Component details

Rock toe and graded filter

The shared general excavation beneath the rock-toe foundation is assigned to the rock-toe trench and deducted from the general cut. It is counted once.



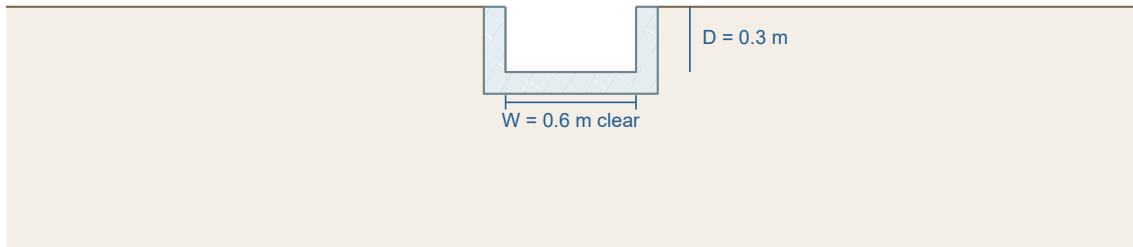
$$\text{Area} = H \times [\text{Crest} + H \times (\text{inner} + \text{outer}) / 2] = 1.2 \times [0 + 1.2 \times (1 + 2) / 2] = 2.16 \text{ m}^2$$

$$\text{Base } 3.6 \text{ m} \cdot \text{inner } 1:1 \cdot \text{outer} = \text{D/S slope}$$

Chute drains

Excavation area: $A = W \times D = 0.6 \times 0.3 = 0.18 \text{ m}^2$. Lined perimeter: $P = W + 2D = 1.2 \text{ m}$. Per chute, excavation = $A \times \text{developed length}$; protection = $P \times \text{developed length} \times \text{lining thickness}$.

CC protection: $t = 0.1 \text{ m}$



Excavation area = $W \times D = 0.6 \times 0.3 = 0.18 \text{ m}^2$

Lined perimeter $P = W + 2D = 0.6 + 2(0.3) = 1.2 \text{ m}$

Cross-section of downstream chute drain

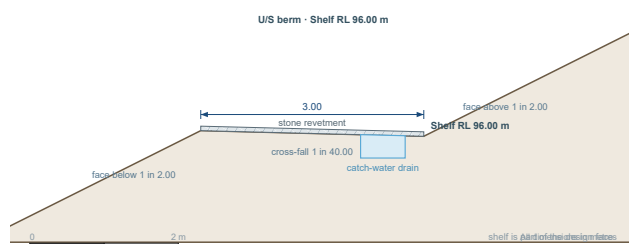
Berm shelves

U/S shelves sit on the left (water face, mirrored); D/S shelves sit on the right. Each figure is labelled with its shelf RL.

U/S berms

U/S berm · Shelf RL 96 m

The shelf is included in the embankment geometry; only its selected surfacing and drain works are additional quantities.

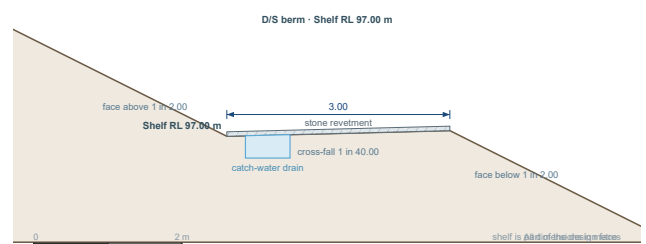


Shelf surfacing: 144 m^2 Drain protection: 7.2 m^3 Drain excavation: 10.8 m^3

D/S berms

D/S berm · Shelf RL 97 m

The shelf is included in the embankment geometry; only its selected surfacing and drain works are additional quantities.



Shelf surfacing: 144 m^2 Drain protection: 7.2 m^3 Drain excavation: 10.8 m^3

Phreatic-line comparison

Reference and selected drainage conditions at the governing cross-section.

Parameter	Without drainage	Selected drainage
Focus datum RL (m)	95	95
Water depth (m)	3.5	3.5
Focal distance S; $q = K \times S$ (m)	0.28	0.277

